

Evaluating Cellular Communication Sensing for Lapse Risk Prediction During Early Recovery from Alcohol Use Disorder

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Alcohol Use Disorder (AUD) is a chronic, relapsing disease. An automated recovery support system using personal sensing and machine learning may help identify when individuals are at elevated lapse risk. Cellular communication sensing may detect dynamic changes in lapse risk and can be contextualized with self-reported, risk-relevant information about contacts. We evaluated a machine learning model predicting next-day alcohol lapse among individuals in early recovery from AUD using contextualized cellular communication data and baseline alcohol use, demographic, and psychiatric and personality characteristics. A total of 144 participants (49% male; mean age=40; 87% non-Hispanic White) with a goal of abstinence provided cellular communication data and alcohol use reports via a 4x daily EMA for up to three months. Models were trained and evaluated using repeated k-fold cross-validation. The best-performing full model used an elastic net algorithm and retained 10 features (median posterior auROC=0.67, 95% Bayesian credible interval (CI; [0.64, 0.71])). A comparison model including only baseline features demonstrated comparable performance (median auROC=0.69, 95% CI [0.65, 0.72])). Cellular communication features on their own performed poorly, but still above chance performance (median auROC=0.59, 95% CI [0.55, 0.62])). These findings demonstrate that cellular communication data capture some risk-relevant signal for alcohol lapse but do not provide incremental predictive value beyond baseline measures. Several communication features were retained in the final model with moderately sized coefficients, suggesting that aspects of social communication may be important for understanding lapse risk. Although, limitations inherent to cellular communication as a sensing method may outweigh their added value.

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Introduction

Alcohol Use Disorder (AUD) is a chronic, relapsing disease [1–3]. Lapses, single episodes of alcohol use, are among the strongest predictors (and a necessary precursor) of relapse, a full return to harmful drinking [4,5]. While lapses can occur at any point in recovery, they are particularly risky during early recovery [6]. Protective coping mechanisms and socio-environmental resources that support recovery are dynamic and accumulate over time [7]. Consequently, early recovery represents a critical window of vulnerability during which a lapse is more likely to escalate into relapse.

Given the dynamic and context-dependent nature of lapse risk, there is a clear need for methods capable of identifying when and why an individual is at increased risk in real time. An automated recovery support system powered by personal sensing and machine learning may assist with this inherently difficult task. Personal sensing of densely sampled data from individuals’ day-to-day lives can provide the inputs necessary for temporally dynamic lapse prediction [8]. Early machine learning models using ecological momentary assessment (EMA) data have achieved excellent accuracy in predicting future lapses back to alcohol use in treatment seeking populations [9–11].

Despite the high predictive success of EMA-based approaches, questions remain about the long-term feasibility of a self-report sensing method. EMA has been shown to be well tolerated among substance-using populations over relatively short periods [12,13]. However, it is unclear whether individuals would be willing or able to adhere to intensive EMA protocols (e.g., four prompts per day) indefinitely. Moreover, EMA items are selected using domain expertise from decades of research on the self-report factors associated with lapse. It is possible, however, that additional lapse precipitants remain undiscovered due to small subtle changes in one’s environment, social circle, or lifestyle that cannot be easily detected via self-report.

Cellular communication sensing may be a promising alternative or complement to EMA. Whereas EMA is limited to, at most, several assessments per day, communication sensing is mostly passive and can be monitored moment-by-moment. Cellular communication patterns capture clear, risk-relevant constructs. Late-night phone calls could indicate an emergency, “drunk dialing,” or interpersonal conflict. A decrease in the number of contacts an individual communicates with could reflect a shrinking social circle, isolation, or disengagement. Furthermore, cellular communication sensing enables data-driven feature engineering, whereby features are systematically derived from raw communication logs and retained based on their predictive utility rather than a priori theoretical assumptions.

These data may become even more powerful when communication patterns are contextualized with participant-specific meaning. Knowing a participant’s relationship to their contacts, whether they have previously drunk alcohol with a given contact, or whether that contact supports their recovery goals could fundamentally alter interpretation. In the examples above, contextualized communication data might reveal that the late-night calls are made to a sponsor, or that a shrinking social circle reflects reduced contact with individuals unsupportive of

their recovery. In this way, the same communication patterns may reflect protective processes rather than increased lapse risk.

This focus on contextualized communication is especially important because lapse is fundamentally social. A substantial body of literature demonstrates that social support and social network influences play a central role in drinking behavior and recovery outcomes [14–16]. Positive social relationships have consistently been associated with better alcohol treatment outcomes [17–20]. For example, the supportive and pro-abstinent social environment of self-help groups such as Alcoholics Anonymous is thought to play a major role in their efficacy [18,21–23]. Conversely, maintaining relationships with heavy drinkers or individuals unsupportive of recovery is associated with poorer outcomes [24–26]. The buffering hypothesis further suggests that positive social relationships mitigate stress by providing resources that promote adaptive responses [27,28], a mechanism that likely extends to lapse risk given stress’s complex relationship with substance use [29].

Importantly, these influences unfold over time through everyday social interactions. Because many social interactions occur over the phone, cellular communication logs provide a natural window into patterns of social engagement. Social relationships are inherently dynamic [30–33], and time-stamped communication data allow for the tracking of how these interactions fluctuate over time.

In this study, we evaluated whether contextualized cellular communication features contain clinically meaningful signals for predicting next-day alcohol lapse risk among individuals in early recovery from AUD. Using a machine learning approach, we examined the predictive utility of these features and identified the most informative communication-based features, with the goal of uncovering novel, clinically relevant predictors of lapse risk.

Methods

Transparency and Openness

We adhere to research transparency principles that are crucial for robust and replicable science. First, we reported how we determined the sample size, all data exclusions, all manipulations, and all study measures. We provide a transparency checklist [34] in the supplement. Second, our features, labels, questionnaires, code, and other study materials are publicly available on our Open Science Framework (OSF) page (<https://osf.io/wgpz9/>).

Participants and Procedure

We recruited 192 adults in early recovery from AUD in Madison, Wisconsin, USA through print and digital advertisements and partnerships with treatment centers from February 15, 2017 through September 19, 2019. This sample size was determined based on traditional

power analysis methods for logistic regression [35] because comparable approaches for machine learning models have not yet been validated. Eligibility criteria required that participants were age 18 or older, able to read and write in English, had moderate to severe AUD (4 self-reported DSM-5 symptoms), had a goal of abstinence from alcohol at the time of the screening visit, had been abstinent for 1–8 weeks, were willing to use a single smartphone, and were not exhibiting severe psychosis or paranoia (defined as scores >2.2 or 2.8 , respectively, on the psychosis or paranoia scales of the Symptom Checklist–90 [36].)

Participants completed up to 5 study visits over approximately 3 months: a screening visit, intake visit, and 3 monthly follow-up visits. At screening we determined eligibility and collected demographic information (age, sex at birth, race, ethnicity, education, marital status, and income) and clinical characteristics (DSM-5 AUD symptom count, personality [37], and psychiatric symptoms [36,38]). At the intake visit, approximately two weeks after screening, we collected additional self-report data on abstinence self-efficacy [39], craving [40], and recent recovery efforts and goals.

At each monthly follow-up, we downloaded backups of participants’ cellular communication metadata directly from their smartphones. Metadata included the phone number of the other party, the date and time of the communication, the origin of call or message (i.e., incoming or outgoing), whether the call was answered (voice calls only), and the duration of the call (voice calls only). During each follow-up visit, study staff identified important contacts. Contacts that participants communicated with at least twice by call or text in the past month were considered important. For each important contact, participants answered seven contextual questions about their type of relationship, whether they ever drank alcohol with this person, the drinking status of the contact, expectations about whether the contact would drink in their presence, recovery status of contact, level of supportiveness of contact, and affective experiences with the contact.

While enrolled, participants completed four brief daily ecological momentary assessments (7–10 questions). The first item assessed alcohol use (date and time of any unreported drinking episodes). The remaining EMA questions were used as features in other studies [10,11], but were outside the scope of this cellular communication sensing study. Additional sensing data streams and self-report measures were collected for the parent grant. We compensated participants up to \$115 per month for completing study tasks (i.e., EMAs, monthly follow-up visits and sharing sensing data) and \$66 per month to offset the cost of their cellphone plan. The full study protocol is available on our OSF page (<https://osf.io/wgpz9/>).

Ethics

All procedures were approved by the University of Wisconsin-Madison Institutional Review Board (Study #2015-0780) and carried out in accordance with the principles of the Declaration of Helsinki. All participants provided written informed consent observed by a research assistant.

Data Analysis Plan

Labels

Our models predicted the probability of an alcohol lapse within a 24-hour window. Predictions were generated daily at 4 a.m., beginning on participants' second study day and continuing for up to 3 months. Participants reported the date and hour of the start and end time of any alcohol use on the first item of the EMA. Participants demonstrated good adherence to our 4X daily EMA protocol (mean adherence = 78%) and provided at least one EMA on 95% of study days. Prediction windows were labeled as lapse if any alcohol use was reported in the 24-hour window. In total, there were 11,507 labeled prediction windows across all participants. Positive lapse labels were underrepresented (7.5%; 861/11,507).

Feature Engineering

Features were calculated using only data collected before the start of each prediction window to ensure our models were making true future predictions. We calculated a total of 181 features from four feature sets:

1. *Cellular Communication Data.* We filtered the raw communication data to include only communications with known context (i.e., people with whom they communicated with at least twice in a month and whom they provided self-report context about). Cellular communication features were engineered from all available data up to the start of each window. We used two feature scoring epochs (24 hours and 1 week before the start of the prediction window) to create a total of 142 features.

Within each feature scoring epoch, we calculated two types of features: raw and difference features. Raw features represent the feature value calculated within a given feature scoring epoch. For example, the raw rate of incoming text messages in a 24-hour feature scoring epoch was calculated as the total number of incoming text messages in the 24 hours immediately preceding the start of the prediction window, divided by 24. Difference features capture participant-level changes from baseline. Specifically, for each feature we subtracted the participant's mean value (using all available data prior to the prediction window) from the associated raw feature value. For example, the difference feature for incoming text messages was calculated as the raw incoming text message rate minus the participant's average incoming text message rate across all time on study.

2. *Demographics.* We created 2 numeric features for age and income, and 10 one-hot coded features for sex at birth (male, female), race and ethnicity (non-Hispanic White, Hispanic and/or not White), education (high school or less, some college, college degree), and marital status (married, not married, other).

3. *Alcohol Use Characteristics.* We created 21 features from baseline measures of alcohol use characteristics, including numeric features of DSM-5 AUD symptom count, craving, four abstinence self-efficacy subscales (negative affect, social, physical, and craving), number of individual alcohol counseling sessions, group alcohol counseling sessions, self-help meetings, and other mental health counseling sessions in the past 30 days, number of days in contact with supportive and unsupportive people in the past 30 days, and satisfaction with and confidence in recovery over the past 30 days. We also created one-hot coded features for whether they have taken prescription medication for alcohol use disorder or mental health in the past 30 days and whether they had a goal of abstinence.
4. *Psychiatric and Personality Characteristics* We created 6 numeric features from baseline measures from broad psychiatric and personality traits, including higher order personality dimensions of positive emotionality, negative emotionality, and constraint and psychiatric comorbidities of depression, anxiety, and stress.

Table 1 details the raw predictors, feature sets, feature engineering procedures, scoring epochs, and number of features. If no data was observed during a feature scoring epoch that feature was scored 0, thus there was no missing data across features. Other feature engineering steps performed during cross-validation included standardizing all features and removing zero and near-zero variance features as determined from held-in data (code for our full data pre-processing pipeline is available on our OSF page [<https://osf.io/wgpz9/>]).

Raw Predictor

Originated
 Call duration
 Call answered
 Date/time of communication
 Phone number
 Type of Relationship
 Have you drank alcohol with this person?
 What is their drinking status?
 Would you expect them to drink in your presence?
 Are they currently in recovery from drugs or alcohol?
 Are they supportive about your recovery goals?
 How are your typical experiences with this person?
 Age
 Sex at birth
 Race
 Education
 Income
 Marital Status
 DSM-5 symptom count
 Craving

Raw Predictor

Abstinence self-efficacy: Negative affect, social, physical, and craving subscales
Number of individual alcohol counseling sessions attended (past 30 days)
Number of group alcohol counseling sessions attended (past 30 days)
Number of self-help group meetings attended (past 30 days)
Number of other mental health counseling sessions attended (past 30 days)
Number of days in contact with supportive people (past 30 days)
Number of days in contact with unsupportive people (past 30 days)
Taken prescribed medication for alcohol use disorder (past 30 days)
Taken prescribed medication for other mental health disorder (past 30 days)
Satisfaction with progress toward recovery goals (past 30 days)
Confidence in abstinence ability (next 30 days)
Has a goal of abstinence
Personality dimensions (positive emotionality, negative emotionality, and constraint)
Psychiatric conditions (depression, anxiety, and stress)

Note:

Cellular communication features were scored over two feature scoring epochs (24 hours and 1 week before the

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Model Selection

We included three candidate statistical algorithms in our model configurations: elastic net, random forest, and XGBoost. These algorithms differ in key ways related to flexibility, complexity, and their ability to accommodate non-linear and interactive relationships between features and lapse probability. Random forest and XGBoost algorithms can natively handle interactions, whereas elastic net assumes an additive data generating process. Thus, if true interactions exist between features, we would expect one of these tree-based methods to yield the best performance. However, if relationships are primarily additive, elastic net may perform better. Model configurations also differed on outcome resampling method (i.e., up-sampling and down-sampling of the outcome at ratios ranging from 5:1 to 1:1), hyperparameter values, and feature set.

We selected three final models. We selected the best performing full model that included all four feature sets (cellular communications, demographics, alcohol use characteristics, and psychiatric/personality characteristics). Because our feature set of interest was cellular communications we only selected configurations that retained this feature set. We selected the best performing baseline model that included our three baseline feature sets (demographics,

alcohol use characteristics, and psychiatric/personality characteristics) to assess the incremental predictive value of cellular communication beyond these baseline measures. Finally, we selected the best performing cellular communication model that included only the cellular communication feature set to evaluate the predictive signal captured in these features.

The best configuration for each of these models was selected using 6 repeats of 5-fold cross-validation. Participants were grouped so that all of their data were always in the held-in or held-out fold for a split, but never in both. Our performance metric was area under the receiver operating curve (auROC). Folds were stratified so that all folds contained comparable proportions of individuals who lapsed frequently (i.e., 10+ times).

Model Evaluation

We evaluated model performance for our three final models (full model, baseline model, cellular communication model) using a Bayesian hierarchical generalized linear model. Posterior distributions with 95% credible intervals (CI) were estimated from the 30 held-out folds using weakly informative, data-dependent priors to regularize and reduce overfitting: Residual SD $\sim$ normal(0, exp(2)); intercept (centered predictors) $\sim$ normal(2.3, 1.3); window-width contrasts $\sim$ normal(0, 2.69); covariance $\sim$ decov(1,1,1,1). Random intercepts were included for repeat and fold (nested within repeat). auROCs were logit-transformed and regressed on model type to estimate the probability that model performances differed systematically. We reported the median posterior auROC and 95% CI for each model. We also specified two sets of contrasts for model comparisons: full model vs. baseline model and cellular communication model vs. a hypothetical model performing at chance (i.e., where auROC's are 0.5 in all 30 test sets).

We refit our full and cellular communication models on the entire data set to identify the most important features. We quantified feature importance by examining the retained features (i.e., coefficient value > 0) in our final elastic net model and ordering them by absolute coefficient value. These values provide an estimate of the direction and magnitude of association between each predictor and the outcome, conditional on the other features retained. We plotted coefficients for both models to visualize whether descriptive differences in retained cellular communication features existed when controlling for other baseline features compared to not controlling for baseline features.

Results

Participants

We enrolled 169 participants. Of these, 151 completed the first follow-up visit where the first cellular communication log download occurred. We excluded data from seven participants due to poor compliance providing communication data (i.e., deleting logs prior to the download or not providing context information about important contacts). The final analytic sample

included 144 participants. Table 2 provides the demographic characterization of our sample. 56% of participants reported at least one lapse while on study.

	N	%	M	SD	Range
Age			40.4	11.8	21-72
Sex at Birth					
Female	74	51.4			
Male	70	48.6			
Race					
American Indian/Alaska Native	3	2.1			
Asian	2	1.4			
Black/African American	8	5.6			
White/Caucasian	125	86.8			
Other/Multiracial	6	4.2			
Hispanic, Latino, or Spanish origin					
Yes	3	2.1			
No	141	97.9			
Education					
Less than high school or GED degree	1	0.7			
High school or GED	14	9.7			
Some college	39	27.1			
2-Year degree	13	9.0			
College degree	55	38.2			
Advanced degree	22	15.3			
Employment					
Employed full-time	70	48.6			
Employed part-time	25	17.4			
Full-time student	7	4.9			
Homemaker	1	0.7			
Disabled	7	4.9			
Retired	8	5.6			
Unemployed	15	10.4			
Temporarily laid off, sick leave, or maternity leave	3	2.1			
Other, not otherwise specified	8	5.6			
Personal Income			\$35,050	\$32,069	\$0-200,000
Marital Status					

	N	%	M	SD	Range
Never married	63	43.8			
Married	32	22.2			
Divorced	42	29.2			
Separated	5	3.5			
Widowed	2	1.4			

Table 2: Demographic Characterization

Note:

N = 144

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Communications

Participants had an average of 26 important contacts (range 2-113) that were contextualized with self-report information. We obtained a total of 375,912 contextualized communications across participants. Participants had, on average, 2,610 contextualized communications (range = 109-14,225) averaging to about 33 communications per day (range 3-278).

Model Evaluation

The median posterior auROC for the best performing full model was 0.67, 95% CI [0.64, 0.7]. The final model fit on the full data set retained 10 features (left panel of Figure 1). The top four were baseline alcohol use characteristics. Two of these (abstinence confidence and having a goal of abstinence) emerged as important protective factors. Abstinence self-efficacy when experiencing negative affect and craving emerged as important risk factors. Communication features were also retained in the final model. Specifically, frequency of communications with drinkers, friends, and people unaware of the individual's recovery goals emerged as important risk factors. Frequency of communications with non-drinkers, conversely, was an important protective factor.

We conducted a model comparison between the full model and a model that only considered baseline measures (alcohol use characteristics, demographics, and psychiatric and personality characteristics) to assess the incremental predictive value of cellular communication features beyond these baseline measures. The baseline model achieved slightly higher performance compared to the full model (median auROC = 0.69, 95% CI [0.66, 0.71]). However, this median difference in auROC was only .01 and the bayesian model comparison showed weak evidence (73% probability) that the baseline model performed better than the full model.

Finally, our best performing model that included only cellular communication features achieved a median auROC of 0.59. The 95% CI ([0.55, 0.62]) did not contain .5. There was strong evidence that this model is performing above chance performance (100% probability) and thus capturing some signal in the data. To better understand which features are capturing this signal, we examined the retained features in the best cellular communication model fit on the full data set. This model retained four features: frequency of communications with friends, people unaware of the individual’s recovery goals, drinkers, and non-drinkers. These four features were also retained in the full model. In the right panel of Figure 1 we display the magnitude of the coefficients for these features when the model does not control for other retained baseline measures.

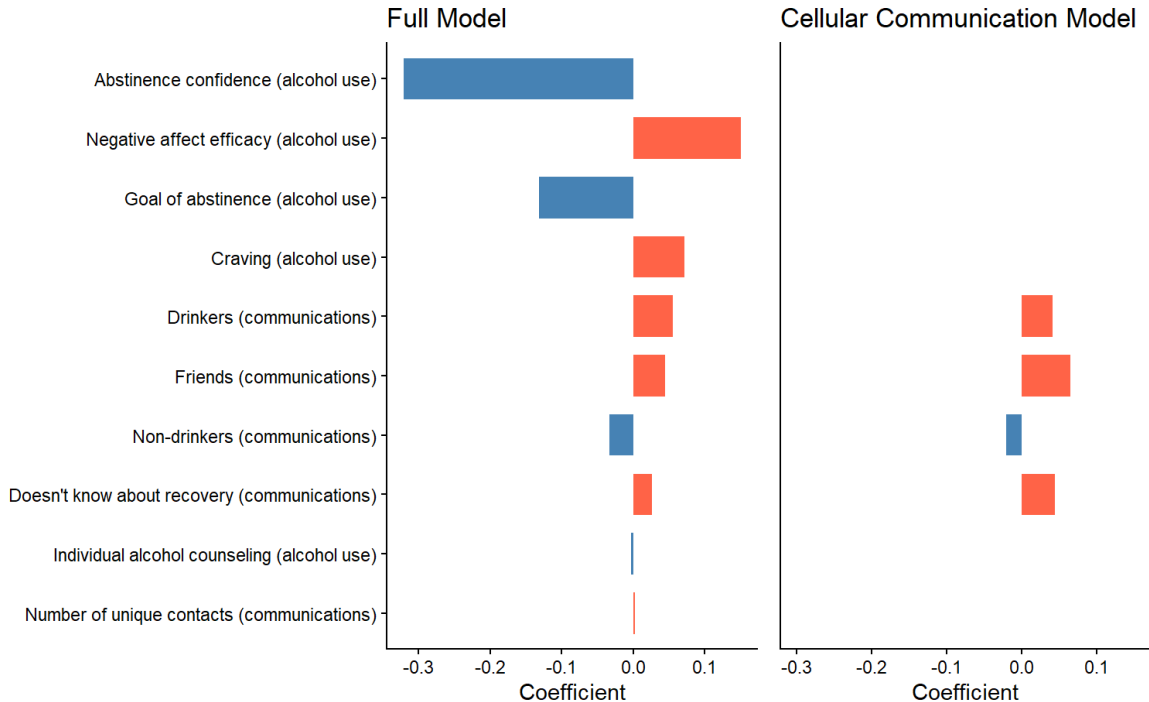


Figure 1: Global feature importance (elastic net coefficient) for the full and cellular communication models. Features are ordered by absolute coefficient value. Bars with positive coefficient values (in blue) represent features that, on average, lower lapse risk. Bars with negative coefficient values (in salmon) represent features that, on average, increase risk. Alcohol use features were collected via self-report measures at baseline. Communication features were engineered from the contextualized cellular communications.

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Discussion

Our model achieved fair performance, with an auROC of 0.67, indicating that some predictive signal was present. However, it did not offer incremental value beyond a baseline model that included only baseline self-report measures. Consistent with this, the four most important predictors in our model were all self-report variables: abstinence confidence, abstinence goal, negative affect efficacy, and craving.

Nonetheless, several communication features were retained in the final model with moderately sized coefficients. These included communications with people unaware of the participant’s recovery status, non-drinkers, friends, and individuals who were unpleasant to interact with. In contrast, raw counts of calls and text messages and call durations were not retained in the final model. This implies that the quantity of communication may be less informative than the quality and social significance. Future research may benefit from collecting richer contextual data about communication contacts to better understand the social dynamics contributing to lapse risk.

Even with highly contextualized communication data, however, prediction may be limited by data sparsity. Many participants had few daily communications, and some had extended periods with no recorded interactions at all. Our study design may have further contributed to this limitation. We collected only phone and SMS text communications through the native smartphone app. In recent years, many individuals use private messaging apps (e.g., WhatsApp, Signal) or social media platforms (e.g., Facebook Messenger, Instagram) as their primary communication method [41]. Therefore, our dataset likely missed a substantial portion of participants’ communications. Future studies could explore whether incorporating communication data from additional platforms yields stronger predictive signal. However, even with improved data collection methods, sparsity may remain a challenge, as some people may simply not communicate frequently with others and others use services (e.g., Snapchat) that automatically delete messages.

We cannot entirely dismiss the potential value of cellular communication data for risk prediction. For example, researchers have successfully incorporated communication data into models with other sensing data (e.g., accelerometer, geolocation, and device usage) to detect current [42] and predict future [43] heavy drinking episodes in non-treatment seeking young adult populations. It is possible that in certain populations cellular communications may hold more signal. Young adults may have more frequent communication reducing sparsity concerns. Additionally, non-treatment seeking populations may be less likely to sensor their data (i.e., deleting communications) when the drinking behavior is not at odds with their goals and/or values. However, even in these instances, the unique contribution of cellular communications beyond other sensing methods is unclear. Some communication features, such as outgoing call duration and the number of outgoing calls emerged in the top 20 important features for detecting current drinking episodes. Conversely, when predicting future drinking episodes, no communication features appeared in the top 20. Other sensing methods, like geolocation and

accelerometer data, appeared to be more robustly important for both detection and prediction.

Other practical challenges in collecting call and text message data further limit the feasibility of this sensing method. For example, we obtained participants' cellular communication data by downloading backups of their communication logs in person during their monthly follow-up visits. It is possible to collect cellular communication data in real time using apps installed on Android devices. However, Apple heavily restricts apps in its app store from accessing call and text message data, making real-time sensing of communications challenging (if not impossible) for IOS users. We conclude that other forms of social interaction characterization (e.g., engineering time spent with supportive contacts from geolocation data) are more worthwhile to pursue in future research.

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Author Contributions

KW contributed to conceptualization, data curation, formal analysis, methodology, visualization, writing – original draft, and writing - review and editing. CY contributed to conceptualization, data curation, methodology, writing - original draft. JJC contributed to conceptualization, data curation, methodology, writing – review and editing, supervision of analysis, funding acquisition.

Declaration of Conflicting Interest

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Data Availability

Our de-identified data, questionnaires, and other study materials are publicly available on our OSF page (<https://osf.io/wgpz9/>).

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